

Approval Status

Approval Status	
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Assumptions/Key Ideas:

We assume the problem can be written as a matrix equation $M \ddot{\theta} + K\theta = \mathbf{0}$, where M is a 2x2 mass matrix and K a 2x1 stiffness matrix. Because the problem has two degrees of freedom, we know that there will be two normal modes, which can be found by solving for the eigenvalues and vectors of MK . This problem could also be solved by starting with the full nonlinear equations for the double pendulum (problem 25), and dropping all $\dot{\theta}$ terms.

Additional Solution:

This problem is also highlighted in

http://www.brown.edu/Departments/Engineering/Courses/En4/Notes/vibrations_mdof/vibrations_mdof.htm which shows both the original equations of motion and the linearized ones.

Solution:

We define a coordinate system for the double pendulum as in Figure 1. We define $\hat{i}$ to be facing downward and $\hat{j}$ to the right, which is different than our typical setup in MAE 5730. This allows us to still define θ as counterclockwise from $\hat{i}$, which is how we most often deal with rotating reference frames.

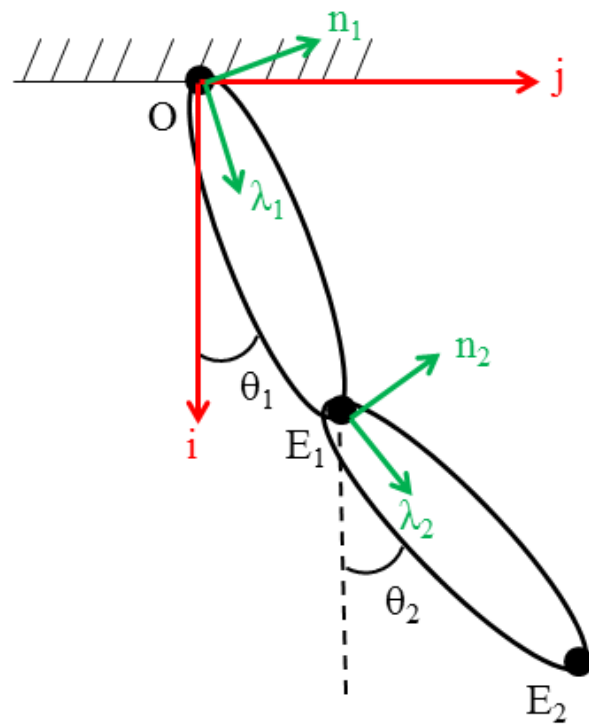


Figure 1: Double Pendulum Setup

The Free Body Diagram for this system is shown in Figure 2. Note that we have already applied some of the simplifications in the problem: the two lengths are the same (l), and all the mass is at the elbow and the hand (which I have labeled as E1 and E2 respectively).

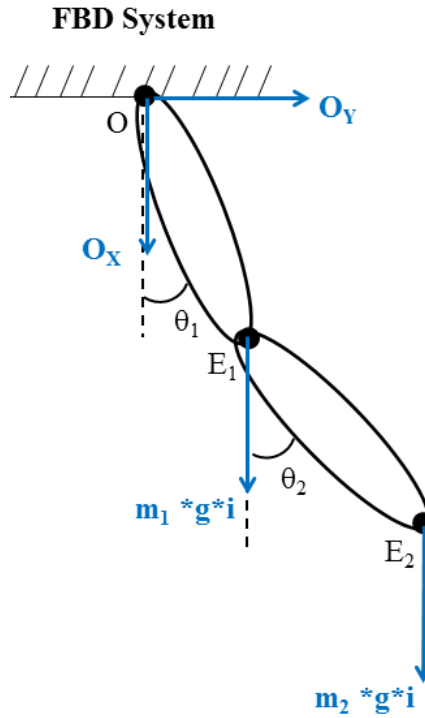


Figure 2: Double Pendulum FBD

Before we can linearize the equations of motion, we first need the general EOM for the problem. Because the only forces acting on the system are gravity and the constraint forces, energy is conserved. This problem is a great candidate for Lagrange Equations. Our generalized Lagrange coordinates q_i will be θ_1 and θ_2 .

We define the necessary position vectors for this problem, making use of our two rotating frames:

Position Vectors	Position Vectors in Inertial Frame
$\mathbf{r}_{e1o} = l\mathbf{\hat{\lambda}}_1$	$\mathbf{r}_{e1o} = l * (\cos(\theta_1) \mathbf{i} + \sin(\theta_1) \mathbf{j})$
$\mathbf{r}_{e2e1} = l\mathbf{\hat{\lambda}}_2$	$\mathbf{r}_{e2e1} = l * (\cos(\theta_2) \mathbf{i} + \sin(\theta_2) \mathbf{j})$
$\mathbf{r}_{e2o} = \mathbf{r}_{e1o} + \mathbf{r}_{e2e1}$	$\mathbf{r}_{e2o} = l * (\cos(\theta_1) \mathbf{i} + \sin(\theta_1) \mathbf{j}) + l * (\cos(\theta_2) \mathbf{i} + \sin(\theta_2) \mathbf{j})$

From here, we can next work on getting the Potential and Kinetic Energy in order to calculate the Lagrangian.

Potential Energy:

$$PE_1 = -mgx_1 = -mgl \cos \theta_1$$

$$PE_2 = -mgx_2 = -mgl (\cos \theta_1 + \cos \theta_2)$$

$$PE_{\text{tot}} = PE_1 + PE_2 = -mgl (2 \cos \theta_1 + \cos \theta_2) \quad *$$

At this point, it is tempting to say $\cos \theta \approx 1$,
but then our $\frac{\partial \mathcal{L}}{\partial \theta_i}$ later will be wrong!

Next, in order to calculate the kinetic energy,
we need the velocity of each mass in the
fixed frame:

For a rotating frame $\vec{v} = \dot{\theta} \hat{k} \times \vec{r}$

$$\vec{v}_1 = \dot{\theta}_1 \hat{k} \times \vec{r}_1 = -l \dot{\theta}_1 \sin \theta_1 \hat{i} + l \dot{\theta}_1 \cos \theta_1 \hat{j}$$

$$\begin{aligned} \vec{v}_2 = \dot{\theta}_2 \hat{k} \times \vec{r}_2 &= (-l \dot{\theta}_1 \sin \theta_1 - l \dot{\theta}_2 \sin \theta_2) \hat{i} \\ &\quad + (l \dot{\theta}_1 \cos \theta_1 + l \dot{\theta}_2 \cos \theta_2) \hat{j} \end{aligned}$$

These are not that nice looking, but we will now
square them (or really dot each with itself):

$$\Rightarrow v_1^2 = l^2 \dot{\theta}_1^2$$

$$v_2^2 = l^2 \dot{\theta}_1^2 + l^2 \dot{\theta}_2^2 + 2l^2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

Here, we can apply the small angle approximation

$$\Rightarrow \cos \theta \approx 1 \Rightarrow v_2^2 = l^2 \dot{\theta}_1^2 + l^2 \dot{\theta}_2^2 + 2l^2 \dot{\theta}_1 \dot{\theta}_2$$

Since we have assumed $I_1 = I_2 = 0$, the kinetic energy for each mass is simply $\frac{1}{2} m v^2$:

$$KE_1 = \frac{1}{2} m l^2 \dot{\phi}_1^2$$

$$KE_2 = \frac{1}{2} m [l^2 \dot{\phi}_1^2 + l^2 \dot{\phi}_2^2 + 2l^2 \dot{\phi}_1 \dot{\phi}_2]$$

$$KE_{\text{tot}} = m l^2 \dot{\phi}_1^2 + \frac{1}{2} m l^2 \dot{\phi}_2^2 + m l^2 \dot{\phi}_1 \dot{\phi}_2$$

With the kinetic & potential energy in hand, we calculate the Lagrangian:

$$\mathcal{L} = KE_{\text{tot}} - PE_{\text{tot}}$$

$$\mathcal{L} = m l^2 \dot{\phi}_1^2 + \frac{1}{2} m l^2 \dot{\phi}_2^2 + m l^2 \dot{\phi}_1 \dot{\phi}_2 + 2mg l \cos \phi_1 + mg l \cos \phi_2$$

We now can get two equations of motion from

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right] - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$$\frac{\partial \mathcal{L}}{\partial q_i} = [-2mg l \sin \phi_1, -mg l \sin \phi_2] \approx [-2mg l \phi_1, -mg l \phi_2]$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = [2m l^2 \dot{\phi}_1 + m l^2 \dot{\phi}_2, m l^2 \dot{\phi}_2 + m l^2 \dot{\phi}_1]$$

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right] = [2m l^2 \ddot{\phi}_1 + m l^2 \ddot{\phi}_2, m l^2 \ddot{\phi}_2 + m l^2 \ddot{\phi}_1]$$

From here, we easily see two equations of motion

$$1) 2ml^2\ddot{\phi}_1 + ml^2\ddot{\phi}_2 + 2mgl\phi_1 = 0$$

$$2) ml^2\ddot{\phi}_1 + ml^2\ddot{\phi}_2 + mgl\phi_2 = 0$$

We divide every term by ml ,
and find $M\ddot{\vec{\phi}} + K\vec{\phi} = \vec{0}$

$$\Rightarrow \begin{bmatrix} 2l & l \\ l & l \end{bmatrix} \begin{bmatrix} \ddot{\phi}_1 \\ \ddot{\phi}_2 \end{bmatrix} + \begin{bmatrix} 2g & 0 \\ 0 & g \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = \vec{0}$$

To find the eigenvalues & eigenvectors
in MATLAB, $[V, D] = \text{eig}(K, M)$;

I attach code that I insert into my
double Pendulum Root as well as
plots of ϕ vs. time showing periodic
motion!

Code

To be inserted into double pendulum simulator just before ode45.

Key takeaway: This section of code is inserted to create the initial conditions used for the pendulum. The square roots of the eigenvalues of $M \backslash K$ are the frequencies of the normal modes, and the eigenvectors are the initial position of the masses. See the full double pendulum code from homework 25 as a reference.

```
% Code for Problem 34, Use Double Pendulum Code to look at normal modes
% Create M and K Matrices
M=[2*1 1; 1 1];
K=[2*g 0; 0 g];
[V,D]=eig(K,M);

%Mode 1
theta1_0=V(1,1);
theta2_0=V(1,2);

%Mode 2
% theta1_0=V(2,1);
% theta2_0=V(2,2);

%Start from rest:
theta1dot_0=0;
theta2dot_0=0;
```

Results

Plots of θ vs. time are (nearly) periodic!

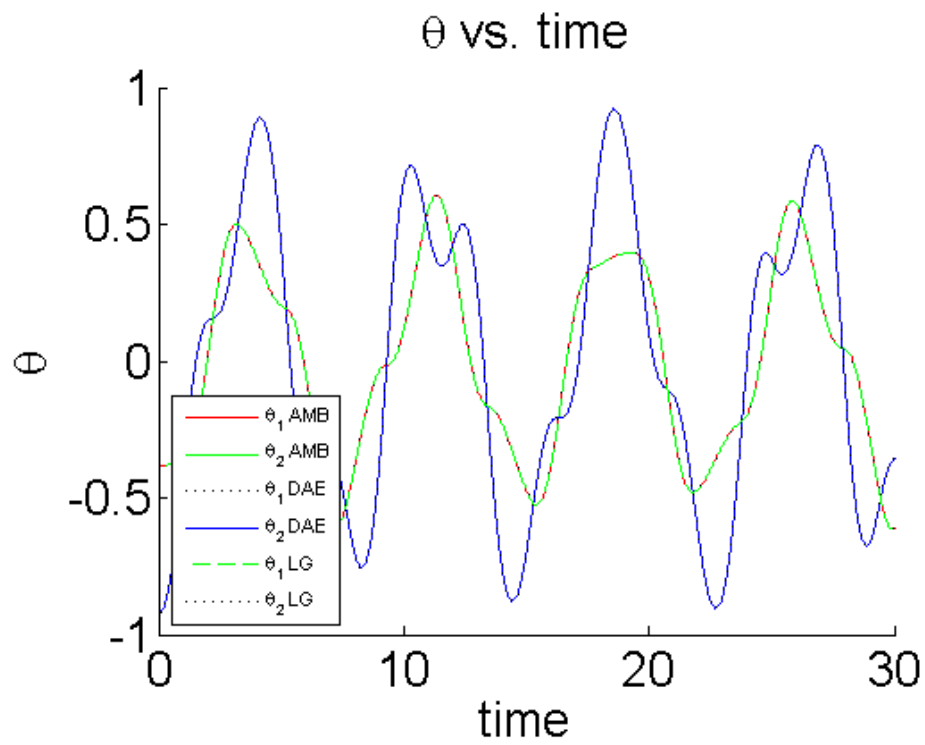


Figure 3: Mode 1 Two Links Swing almost like a Single Pendulum

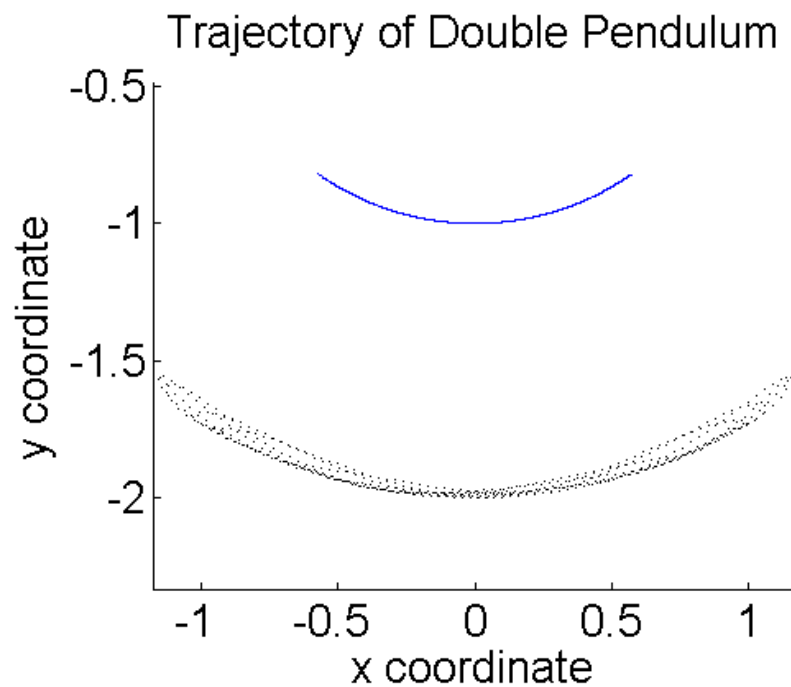


Figure 4: Trajectory of Mode 1

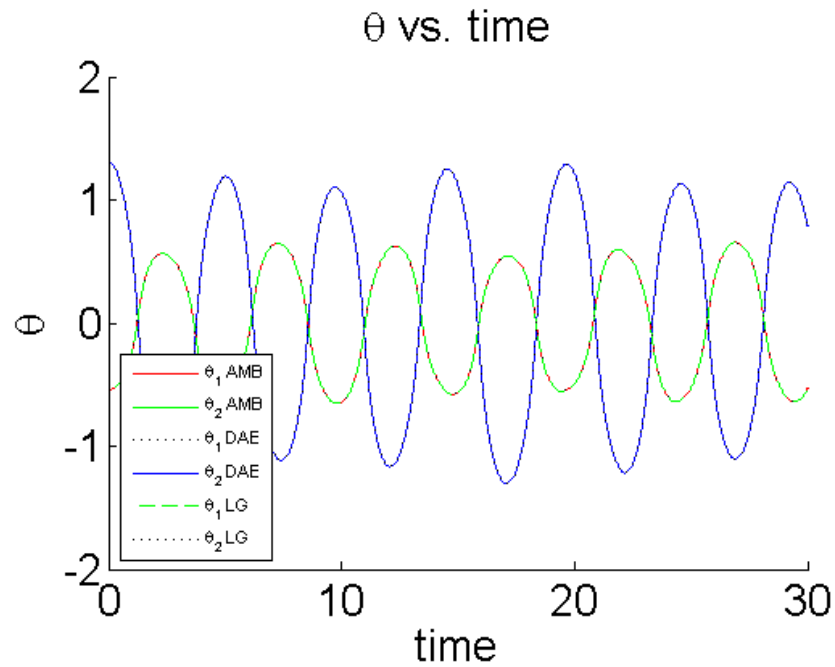


Figure 5: Mode 2: Two Links swing opposite of each other

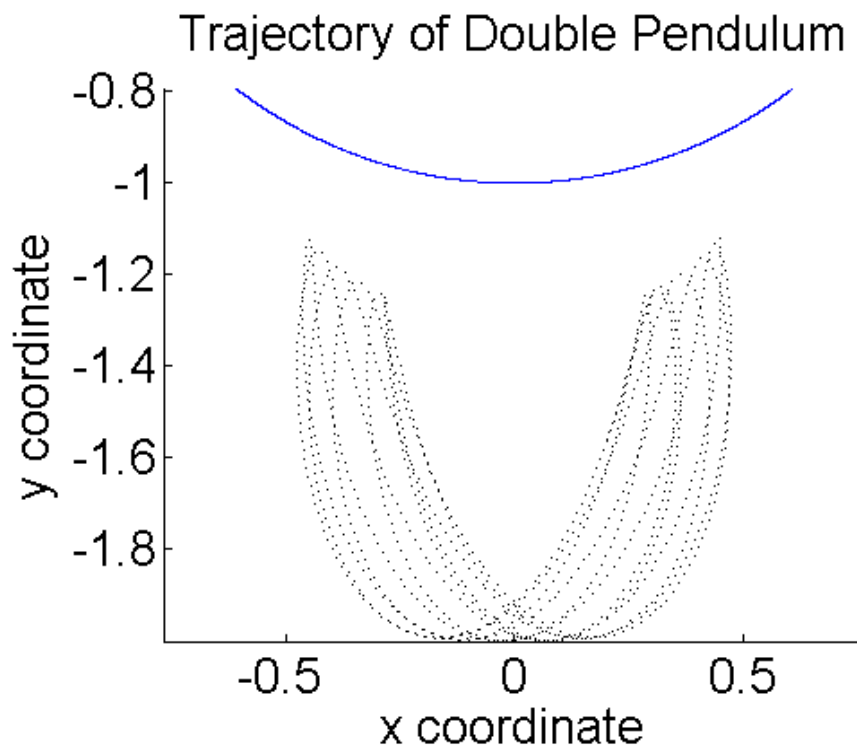


Figure 6: Trajectory of Mode 2